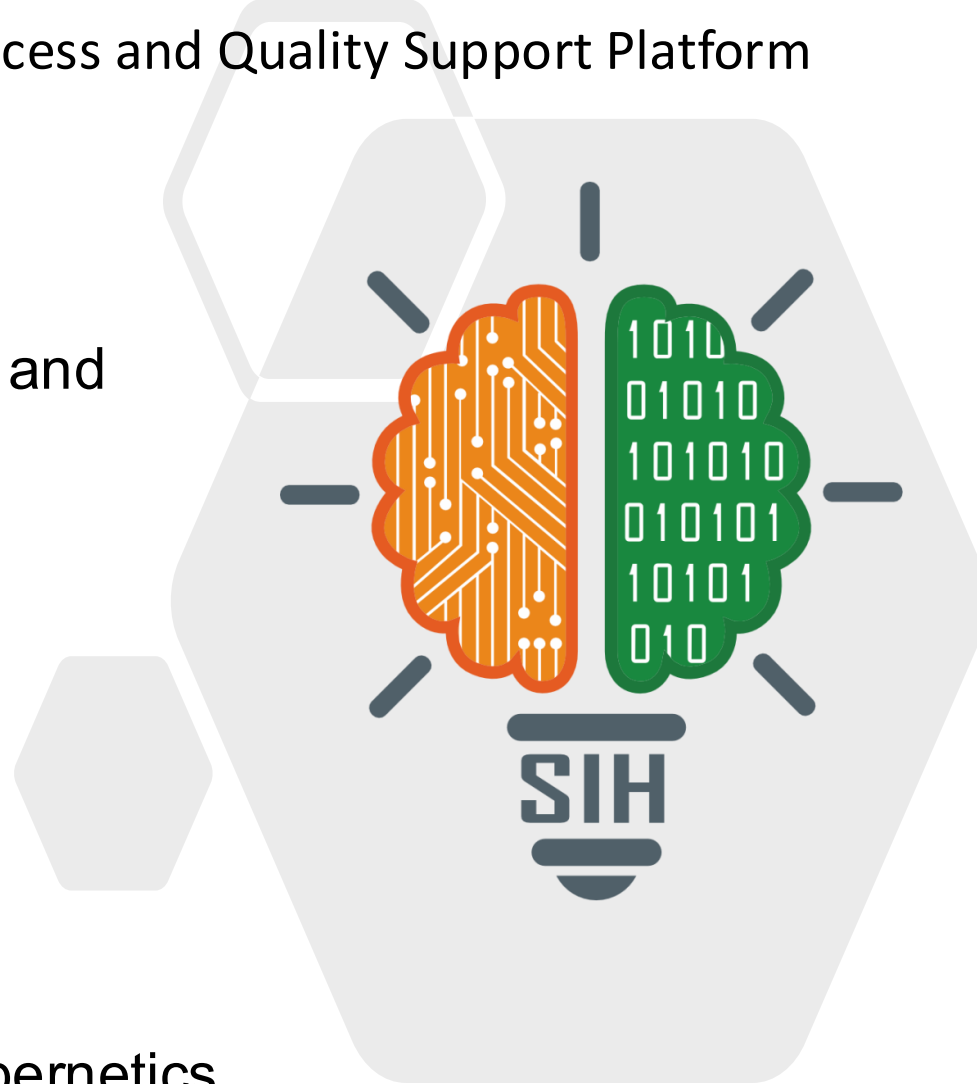


SMART INDIA HACKATHON 2026



SevaSetu: Integrated Care-Access and Quality Support Platform

- **Problem Statement ID** – 26133
- **Problem Statement Title** – Accessibility and quality of public healthcare services.
- **Theme** - HealthTech
- **PS Category**- Software
- **Team ID** - SIH26133
- **Team Name(Registered on portal):** Cybernetics



SevaSetu

Integrated Care-Access and Quality Support Platform

Detailed explanation of the proposed solution:

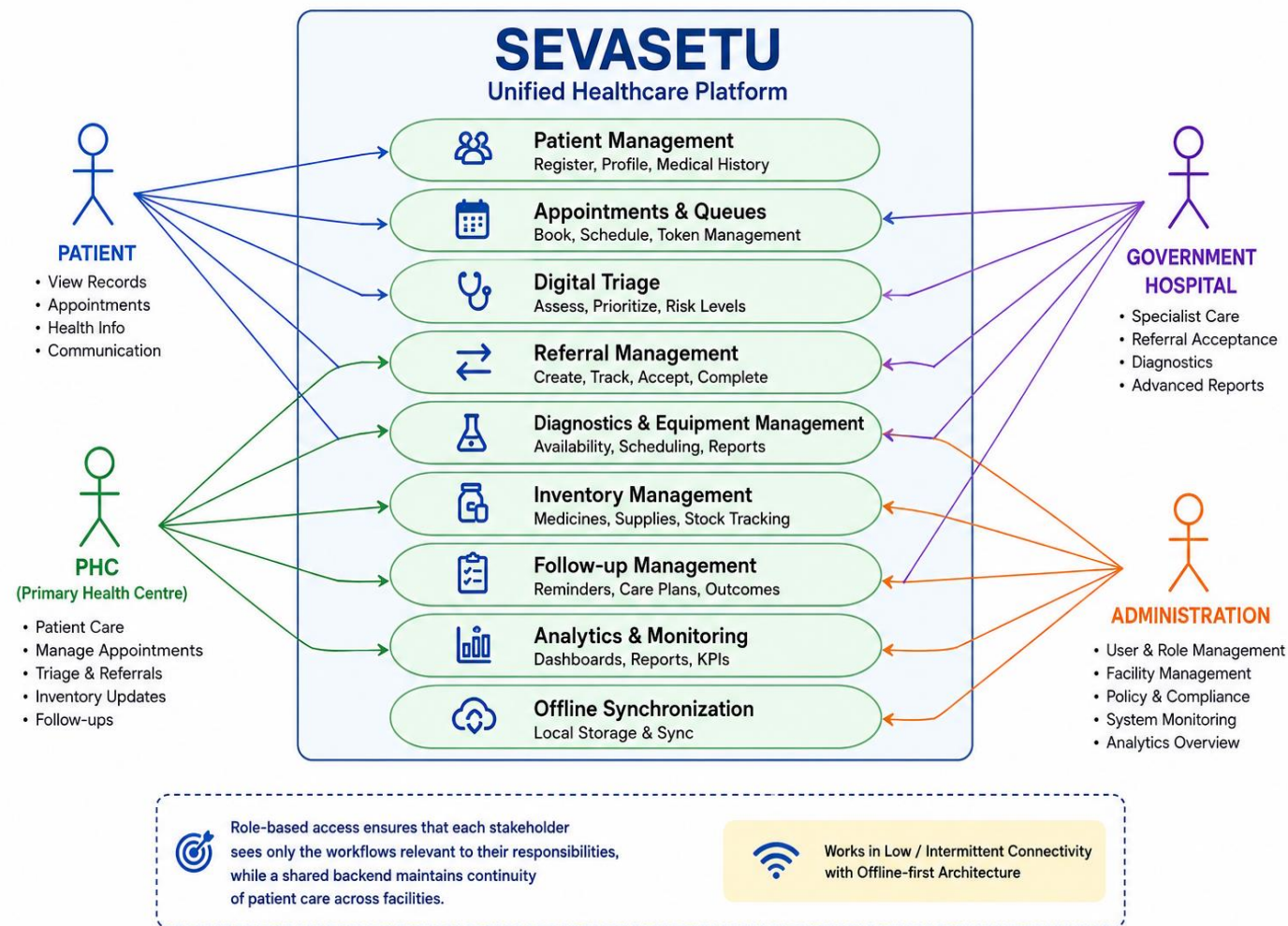
- Patient/ASHA offline-first PWA — booking, digital triage, teleconsultation, multilingual UI. **[Hackathon MVP]**
- Doctor/PHC dashboard — live queue, longitudinal records, teleconsult console, one-click referrals. **[Hackathon MVP]**
- District admin dashboard — facility map, inventory tracking, high-risk follow-up, analytics.

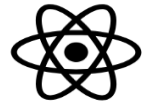
How it addresses the problem:

- Unified FHIR-based record ends fragmentation across sub-centre → PHC → district hospital
- Teleconsultation + facility-matcher closes travel-distance and delayed-referral gaps
- Offline-first design + multilingual UI solves connectivity and health-literacy barriers.

Innovation and uniqueness:

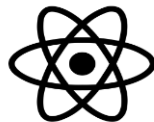
- **Store & Forward Tele-health:** Captures patient data completely offline and auto-syncs to doctors when the network returns.
- **Smart Facility Routing:** Checks live doctor and equipment availability *before* creating a referral to prevent wasted travel.
- **Offline Triage & ABDM Native:** Calculates patient risk (Red/Yellow/Green) locally on device and integrates smoothly with national ABHA IDs.





Client Layer

An Offline-First Progressive Web App built for resource-constrained environments. Utilizes local browser APIs to cache UI shells and patient inputs securely.



React



Vite



Indexed DB



Core Services

A robust, microservices-inspired backend. Handles complex state machines for referrals, teleconsultation routing, and role-based access control (RBAC).



SpringBoot

Java



JWT Auth



Data & Caching

Relational storage for permanent health records, coupled with in-memory caching to achieve O(1) time complexity for live hospital availability checks.



PostGRE SQL

Redis



Apache Kafka

RISKS AND MITIGATION:

- **AI triage errors**
AI used for decision support only, never autonomous diagnosis.
- **Data privacy and security**
RBAC, end-to-end encryption, secure authentication.
- **Offline sync conflicts**
Timestamped, versioned syncs with unique operation IDs.
- **Low user adoption**
Multilingual, role-specific interfaces cut the learning curve.
- **Interoperability**
Standards-based APIs for integration with govt. health systems.

Viability of the solution:

- **Cost-effective:** open-source stack and lightweight infra cut deployment cost by an estimated 60–70% vs. proprietary systems.
- **Simple adoption:** multilingual, role-based UI minimizes training time for ASHA and frontline workers
- **Lower operational load:** digitized appointments, referrals, queues and follow-ups cut manual admin work
- **Reaches low-connectivity areas:** offline-first design keeps essential services running without reliable internet.

Parameter	Existing Solutions (e.g., eSanjeevani)	SevaSetu (Proposed)
Setup Cost	High (Requires steady internet/hardware)	Low (Runs on basic smartphones)
Connectivity	Live Video (Fails in 2G)	Offline-First (Store & Forward)
Scalability	Manual routing	Automated Smart Facility Routing

Reduced travel and waiting



Patients view appointments, referrals and diagnostic queues before visiting facilities.

Improved referral completion



End-to-end referral tracking reduces delays and missed referrals.

Continuity of care



Longitudinal records keep patient information accessible across PHCs and government hospitals.

Better resource utilization



Real-time visibility of medicines, equipment, queues and workforce improves facility management.

Faster triage and care



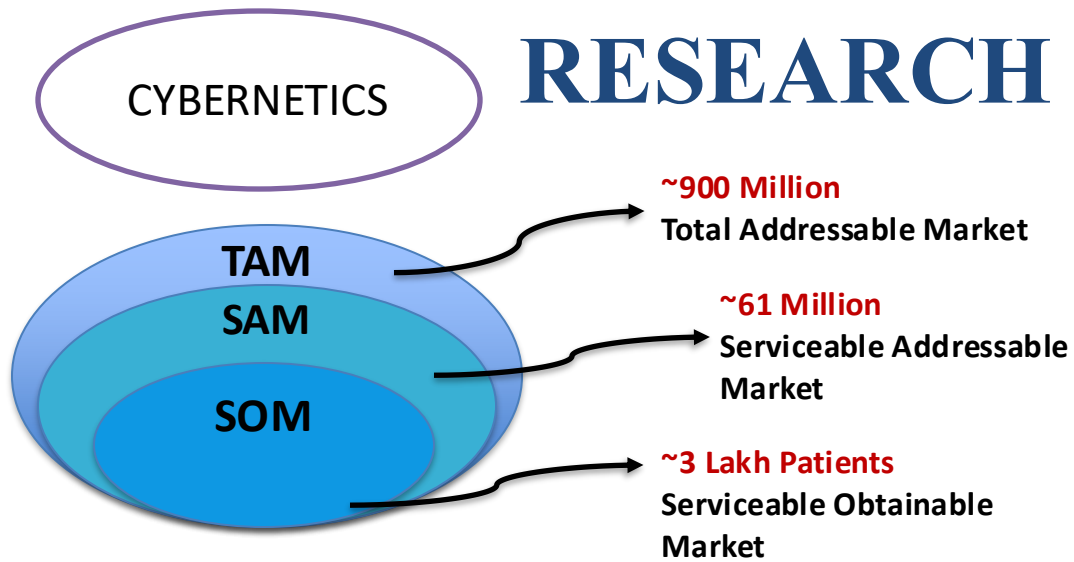
AI-assisted triage helps prioritize high-risk cases and guide appropriate care pathways.

Improved accessibility



Offline-first operation extends reliable healthcare support to low-connectivity regions.

RESEARCH AND REFERENCES



TAM-Total Addressable Market (Indian Rural Healthcare)

The entire Indian rural population facing severe doctor shortages, connectivity issues, and fragmented public healthcare access.

SAM-Serviceable Addressable Market (Maharashtra State Focus)

Targeting Maharashtra's rural population currently dependent on primary health centers (PHCs) and sub-centers for initial diagnosis.

SOM-Serviceable Obtainable Market (Year 1 Initial Adoption)

Pilot rollout across 5 underserved districts. We expect to reach 3 Lakh+ patients via ASHA workers assuming a 5% early adoption rate.



REFERENCE

- **Government Benchmarks:** eSanjeevani (Hub-and-spoke model, but heavily dependent on live video) & **ABDM Sandbox** (National digital health backbone & ABHA IDs).
- **Private Competitors Analyzed:** Practo / Tata 1mg (Urban/semi-urban focus) and Tattvan e-Clinic (Closest rural franchise competitor, but lacks offline-first autonomy).
- **Technical Standards & Links:**
 - **HL7 FHIR Specification:** For interoperable health records (hl7.org/fhir).
 - **W3C PWA Standards:** For Offline-first sync & IndexedDB storage.
 - **Web Demo:** <https://shrinidhi1403.github.io/SevaSetu-SIH-2026/>



LINKS

🔗 IMPORTANT RESEARCH & REFERENCE LINKS

- **ABDM API Sandbox (Govt Integration):** sandbox.abdm.gov.in
- **ABDM FHIR Profiles (India Standards):** nrces.in/ndhm/fhir/r4/
- **eSanjeevani (Market Benchmark):** esanjeevani.mohfw.gov.in
- **W3C PWA Offline Specs:** developer.mozilla.org/en-US/docs/Web/Progressive_web_apps
- **SevaSetu Prototype Code:** <https://shrinidhi1403.github.io/SevaSetu-SIH-2026/>